

TPC-409: An Odd Complete-Shell Height Ladder

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Abstract

TPC-408 extended a locally normalized adjacent-entry proxy to two odd complete prime shells. We now fix the first of those shells, $Q = 65536$, and vary the kernel height through $H = 16, 32, 66, 128$, always with $N = 4H$. All 5709 shell primes are retained, giving the explicit parity counts $m_- = 2854$ and $m_+ = 2855$. The exact local identity and Cauchy–Schwarz proof give, at every height, $0 \leq z \leq t_1/(a_{\min}\sqrt{S_0S_1}) \leq 4/(a_{\min}H) \leq 4/H$. An exact rational certificate and independent literal CRT masked-energy replay verify all four rows. This remains a finite synthetic proxy result, with no arithmetic or twin-prime conclusion.

1 Odd-shell height ladder

Fix $Q = 65536$. The complete shell $(Q, 2Q]$ contains $r = 5709$ primes $p_0 < \dots < p_{r-1}$. For each $H \in \{16, 32, 66, 128\}$ set $N = 4H$ and choose a CRT representative above 10^6 satisfying

$$o \equiv 0 \pmod{p_i} \quad (i \text{ even}), \quad o \equiv -N \pmod{p_i} \quad (i \text{ odd}).$$

All shell primes are retained. Thus $m_- = \lfloor r/2 \rfloor = 2854$ and $m_+ = \lceil r/2 \rceil = 2855$. Define

$$t_d = \frac{H^2}{H^2 + d^2}, \quad S_0 = \sum_{d=1}^{N-1} t_d^2, \quad S_1 = \sum_{d=1}^{N-2} t_d^2 + t_1^2, \quad a_i = \frac{p_i^3}{Q^2(p_i - 1)}.$$

For the odd/even amplitude sums P_-, V_-, V_+ , the TPC-404 local proxy is

$$G_0 = V_- S_0, \quad G_1 = V_- S_1 + V_+(S_1 - t_1^2), \quad M = t_1 P_-.$$

The height H is a proxy parameter, not the physical h_0 .

2 Finite height theorem

Theorem. With $a_{\min} = \min_i a_i$ and $z = M/\sqrt{G_0 G_1}$,

$$0 \leq z \leq \frac{t_1}{a_{\min}\sqrt{S_0 S_1}} \leq \frac{4}{a_{\min}H} \leq \frac{4}{H}. \quad (1)$$

Proof. Since $p_i > Q > N$, the even class masks offset zero and the odd class first masks offset N ; no prime masks offset one. Hence the displayed local identities hold. Cauchy–Schwarz and the odd-class lower bound give $P_-^2 \leq m_- V_-$ and $V_- \geq m_- a_{\min}^2$, while $G_1 \geq V_- S_1$. Consequently

$$z^2 \leq \frac{t_1^2 m_-}{V_- S_0 S_1} \leq \frac{t_1^2}{a_{\min}^2 S_0 S_1}.$$

For $1 \leq d \leq H$, $t_d \geq 1/2$ and those terms occur in both sums, so $S_0, S_1 \geq H/4$. Also $a_i = (p_i/Q)^2 p_i / (p_i - 1) > 1$. Taking square roots proves (1). $\square$

3 Exact certificate and observations

The exact producer enumerates the full shell and uses integer CRT and rational arithmetic. The resulting finite observations are

H	N	m_-	m_+	z	$H z$
16	64	2854	2855	0.021772983081669	0.348367729306709
32	128	2854	2855	0.010803169700743	0.345701430423767
66	264	2854	2855	0.005213967369619	0.344121846394880
128	512	2854	2855	0.002682449831865	0.343353578478674

The table contains float64 observations extracted from exact rational squares; it is not an asymptotic fit. The independent checker reconstructs the sieve and CRT for each height and literally visits every per-prime, per-coordinate mask before comparing both row energies and M .

4 Route boundary

This is a finite four-height result for one synthetic adjacent normalized proxy entry. It is not a full normalized operator estimate, physical h_0 theorem, arithmetic sign or L^2 theorem, fixed-power saving, Route-A or Route-B closure, or twin-prime result. The exact certificate status is `PROVED_EXACT_FINITE_ODD_COMPLETE_SHELL_HEIGHT_LADDER`; arithmetic advance is `NO`, fixed-power credit is `0`, and full Gate B is `OPEN`.

Reproduction

Run the producer, independent replay, and nine-mutation stress checker with `--check` in normal and optimized Python modes. Bridge-B repeats all checks, requires empty stderr and identical output, and locks the release artifacts by SHA-256.